Bruno Henrique Nunes

Data Scientist | Machine Learning & Computer Vision

Brazil | Open to remote opportunities hiring from Brazil

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Professional Summary

Data Scientist and Applied Machine Learning Engineer with a completed MSc in Artificial Intelligence from UNICAMP. Apply Python, SQL and Apache Spark to statistical analysis, data quality, NLP and reproducible machine learning. Experience includes experiment design, model evaluation, semantic search, topic modeling, computer vision and enterprise analytics.

Technical Skills

Data Science and analytics: Python, SQL, Apache Spark, statistical analysis, experiment design, data profiling, data quality, schema validation, metric design.

NLP and Applied AI: Natural Language Processing (NLP), text classification, semantic similarity, text embeddings, vector search, Information Retrieval, semantic search, topic modeling, clustering, RAG, AI agents, LLM evaluation.

Machine learning: PyTorch, computer vision, deep learning, segmentation, model evaluation, ensemble inference, data provenance and reproducibility.

Software engineering: FastAPI, REST APIs, automated testing, observability, AWS Lambda, AWS SES, Git, Azure DevOps CI/CD.

Selected Research and Projects

Deep Learning for Prostate Histopathology Feb 2024 – Sep 2026

Master's research, University of Campinas (UNICAMP)

- Developed a research pipeline for binary segmentation of prostate cancer in histopathological images, using CNN/Transformer-based modeling and PyTorch.
- Implemented data preparation and quality control, patient-level data partitioning, model training, ensemble optimization and final evaluation.
- Preserved configurations, data provenance and experimental outputs to support reproducibility. The REDU deposit contains generated research artifacts; original DIAGSET images are not redistributed.

Code: github.com/bhnunes/Master_Coding | Artifacts: doi.org/10.25824/redu/EDPCF2

GEODITE — Equivariant Interatomic Potentials

Contributed to model implementation, training and evaluation for an equivariant message-passing model for interatomic potentials.

Co-author, *Equivariant Interatomic Potentials without Tensor Products* (arXiv preprint, January 2026).

arxiv.org/abs/2601.15492

Professional Experience

Application Developer — IBM Aug 2026 – Present

- Apply Python, SQL and statistical methods to structured data workflows, with attention to data validation, analytics readiness and reproducible reporting.
- Contribute to AI agent and LLM applications using RAG, NLP and evidence retrieval; build Python/FastAPI services and REST APIs.
- Implement prompt engineering, prompt-injection defenses, output guardrails, automated testing and observability for reliable AI applications.

Selected 2026 Applied AI and Data Science Projects

- **LLM response quality (Feb–Mar):** Combined lexical metrics, semantic similarity, NLP and text embeddings with model-assisted judgment for ambiguous cases.
- **Agent efficiency and tooling (May–Jul):** Designed paired analyses using the Wilcoxon signed-rank test, bootstrap confidence intervals and token usage analysis; built Python tooling for multi-agent workflows, automated testing and code-quality checks.
- **Data Science and analytics (Jul–Oct):** Prototyped RAG-supported, hypothesis-driven analysis and reproducible reporting; applied data quality, NLP semantic search, topic modeling, statistical metric design, Apache Spark and telemetry enrichment with REST APIs, PostgreSQL and Kubernetes.

Teaching Experience

Teaching Assistant — UNICAMP Jul 2024 – Dec 2025

Supported more than 200 undergraduate students in database management and object-oriented programming; led problem-solving sessions in SQL, relational algebra and OOP and supported assessment. Internship appointment.

Selected Earlier Experience

Software Developer — PAteam (freelance) Jan 2024 – May 2024

Built Python data-ingestion automation using AWS Lambda, REST APIs and AWS SES.

Software Developer — Expleo Group (contract) May 2022 – Aug 2023

- Built Python/Robocorp automation and API integrations for a legacy-to-Salesforce migration using Salesforce REST APIs and SQL Server; documented workflows and supported Azure DevOps CI/CD, monitoring and issue resolution.

Senior RPA Developer — Nestlé Feb 2022 – Aug 2023

- Delivered 13 enterprise automations using Python, Blue Prism, PowerShell, SQL and Automation Anywhere; reviewed code across 10+ projects, mentored developers and supported change approval for more than 50 robots.

Senior RPA Developer — Deloitte Nov 2021 – Feb 2022

Translated stakeholder requirements into Python, Node.js and UiPath automation for a healthcare IoT integration project; presented solutions to clients and suppliers.

IBM — Earlier roles

RPA Developer: Oct 2019 – Nov 2021 | Software Developer: Oct 2018 – Oct 2019

IT Analyst: Jan 2017 – Oct 2018 | Support Specialist: Jan 2012 – Oct 2014

- Delivered automations and integrations with Python/Selenium, APIs, SAP, Db2, Cloudant, Blue Prism, VBA and JavaScript.
- Created invoice analytics with VBA and Power Query for 20,000+ records and 10+ validation KPIs; reported processing time fell from more than three hours to about six minutes.
- As Support Specialist, provided English-language support to US and Canadian clients and used SQL for compliance checks and billing-adjustment reporting.

Education

MSc in Artificial Intelligence — University of Campinas (UNICAMP) 2024 – 2026

Completed September 2026; research in deep learning for prostate histopathology.

MBA in Computer Software Engineering — FIA Business School 2022 – 2023

Bachelor's in Engineering Physics / Applied Physics — University of São Paulo 2006 – 2010

Selected Certifications

AWS Certified Generative AI Developer – Professional Oct 2026

AWS Certified Machine Learning Engineer – Associate Jul 2026

AWS Certified Developer – Associate Oct 2025

AWS Certified Solutions Architect – Associate Apr 2025

Languages

Portuguese: Native. **English:** Full professional proficiency; TOEIC 890 (Sep 2025).